

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

06

WAVE



WAVE

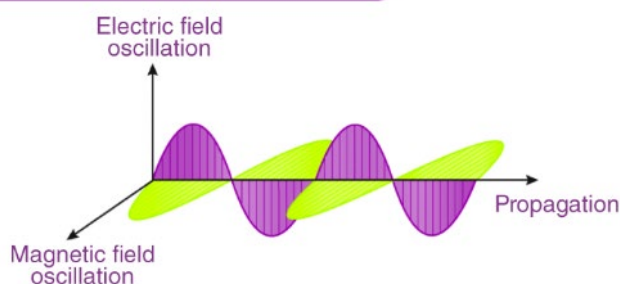
Wave is a disturbance that transfers energy from one place to another without transfer of matter

Example: Sound waves, water waves, light waves

Electromagnetic Waves

- Electromagnetic (EM) waves are a form of energy consisting of oscillating electric and magnetic fields that can travel through a vacuum at the speed of light.
- They are created by the acceleration of charged particles and travel as transverse waves, with the electric and magnetic fields perpendicular to each other and to the direction of travel.

ELECTROMAGNETIC WAVES



Electromagnetic waves are shown by a sinusoidal graph. It consists of time-varying electric and magnetic fields which are perpendicular to each other and are also perpendicular to the direction of propagation of waves. Electromagnetic waves are transverse in nature. The highest point of the wave is known as the crest while the lowest point is known as a trough. In vacuum, the waves travel at a constant velocity of $3 \times 10^8 \text{ m.s}^{-1}$.

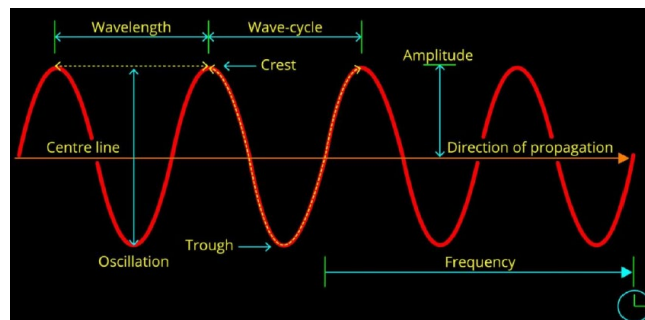
Terms related to Electromagnetic waves

Wavelength: The distance between successive crests of a wave.

Frequency: Wave cycle passing a given point in a given period of time.

Amplitude: Measured between the crest and the mid-point of a wave.

Velocity : Speed and direction of propagation.



Features of Electromagnetic Waves

1. No Medium Required

- Travel in vacuum (space)
- Unlike sound waves

2. Transverse Nature

- Electric field (E) $\perp$ Magnetic field (B) $\perp$ Direction of propagation

3. Speed of EM Waves

- Same for all EM waves in vacuum
- Independence of frequency

4. Produced by Accelerating Charges

- Moving charged particles generate EM waves

5. Carry Energy and Momentum

- Can exert pressure (Radiation pressure)
- Example:** Solar radiation

6. Obey Wave Phenomena

- Reflection
- Diffraction
- Interference

7. Do Not Need Material Medium

- Can travel through empty space (space communication possible)

8. Electric & Magnetic Fields Oscillate

- Fields vary sinusoidally
- Always in phase

9. Relationship Between Speed, Frequency & Wavelength

10. Wide Range of Spectrum

- Includes radio waves to gamma rays
- Differ only in wavelength & frequency



**Different kind of Electromagnetic Waves and their applications**

S.	Radiation	Discover	How produced	Wavelength Range	Frequency range	Energy range	Properties	Application
1	γ -Rays	Henry Becquerel and Madame Curie	Due to decay of radioactive nuclei.	10^{-14} to 10^{-10} m	3×10^{22} Hz to 3×10^{15} Hz	10^6 eV to 10^9 eV	(a) High penetrating (b) Uncharged (c) Low ionising power	(a) gives information on Nuclear structure (b) Medical treatment etc.
2	X-Ray	Roentgen	Due to collisions of high energy electrons with heavy target.	6×10^{-12} m to 10^{-9} m	5×10^{19} Hz to 3×10^{17} Hz	2.4×10^5 eV to 1.2×10^6 eV	(a) Low Penetrating power (b) other properties similar to γ -rays except wavelength	(a) Medical diagnosis and treatment (b) Study of crystal structure (c) Industrial radiography
3	Ultraviolet Rays	Ritter	By ionised gases, sun lamp spark etc.	6×10^{-10} m to 3.8×10^{-7} m	5×10^{17} Hz to 7×10^{14} Hz	2×10^3 eV to 3 eV	(a) All properties of light (b) Photoelectric effect	(a) To detect adulteration, writing and signature (b) Sterilization of water due to its destructive action on bacteria
4	Visible light Subparts of visible spectrum (b) Blue (c) Green (d) Yellow (f) Red	Newton	Outer orbit electron transitions in atoms, gas discharge tube, incandescent solids and liquids.	3.8×10^{-7} m to 7.8×10^{-7} m 3.9×10^{-7} m to 4.55×10^{-7} m 4.55×10^{-7} m to 4.92×10^{-7} m 4.92×10^{-7} m to 5.77×10^{-7} m 5.77×10^{-7} m to 5.97×10^{-7} m 5.97×10^{-7} m to 6.22×10^{-7} m 6.22×10^{-7} m to 7.80×10^{-7} m	8×10^{14} Hz to 4×10^{14} Hz 7.69×10^{14} Hz to 6.59×10^{14} Hz 6.59×10^{14} Hz to 6.10×10^{14} Hz 6.10×10^{14} Hz to 5.20×10^{14} Hz 5.20×10^{14} Hz to 5.03×10^{14} Hz 5.03×10^{14} Hz to 4.82×10^{14} Hz 4.82×10^{14} Hz to 3.84×10^{14} Hz	3.2 eV to 1.6 eV	(a) Sensitive to human eye	(a) To see objects (b) To study molecular structure
5	Infra-Red waves	William Herschell	(a) Rearrangement outer orbital electrons in atoms and molecules. (b) Change E of molecular vibrational and rotational energies (c) By bodies at high temperature.	7.8×10^{-7} m to 10^{-3} m	4×10^{14} Hz to 3×10^{11} Hz	1.6 eV to 10^{-3} eV	(a) Thermal effect (b) All properties similar to those of light except λ	(a) Used in industry, medicine and astronomy (b) Used for fog or haze photography (c) Elucidating molecular structure.
6	microwaves	Hertz	Special electronic devices such as klystron tube	10^{-3} to 0.3 m	3×10^{11} Hz to 10^9 Hz	10^{-3} eV to 10^{-5} eV	(a) Phenomena of reflection, refraction and diffraction	(a) Radar and telecommunication. (b) Analysis of fine details of molecular structure
7	Radio waves Subparts of Radio spectrum	Marconi	Oscillating circuits	0.3 to few kms.	10^9 Hz to few Hz	10^{-3} eV to -0	(a) Exhibit waves like properties more than particle like properties	(a) Radio communication
(A)	Super High Frequency (a) SHF Ultra High Frequency (b) UHF Very High Frequency (c) VHF			0.01 m to 0.1m 0.1 m to 1m 1 m to 10 m	3×10^{10} Hz to 3×10^9 Hz 3×10^9 Hz to 3×10^8 Hz 3×10^8 Hz to 3×10^7 Hz		Radar, Radio and satellite communication (Microwaves), Radar and Television broadcast short distance communication, Television communication.	
(B)	High Frequency (HF) Medium Frequency (MF) Low Frequency (LF) Very Low Frequency (VLF)			10m to 100 m 100 m to 1000 m 1000 m to 10000 m 10000 m to 30000 m	3×10^7 Hz to 3×10^6 Hz 3×10^6 Hz to 3×10^5 Hz 3×10^5 Hz to 3×10^4 Hz 3×10^4 Hz to 10^4 Hz		Medium distance communication Telephone communication, Marine and navigation use, long range communication. Long distance communication.	

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